

WUNSTORF, Germany

WMO: 103340

Lat: 52.4606N

Lon: 9.4311E

Elev: 227

StdP: 14.58

Time Zone: 1.00 (EUC)

Period: 94-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	15.4	19.6	10.4	9.4	17.6	14.3	11.4	21.4	28.5	45.5	25.5	44.3	8.3	80	0.559

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	18.0	87.6	67.3	83.3	65.9	79.6	64.4	69.1	82.9	67.4	79.8	65.8	76.6	8.7	120

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
64.6	91.9	72.5	62.9	86.4	71.6	61.2	81.5	70.4	33.2	83.3	31.9	80.1	30.7	76.8	76.8

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	DB	WB
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(4) 22.9	20.2	18.0	10.7	94.0	6.2	3.2	6.2	96.3	2.6	98.2	-0.9	100.0	-5.4	102.4		
(5)			10.6	72.6	6.4	1.8	6.0	73.9	2.3	75.0	-1.3	76.0	-5.9	77.3		

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
Temperatures, Degree-Days and Degree-Hours	DBAvg	50.6	35.5	37.0	42.0	49.7	56.7	62.3	66.6	66.0	59.1	51.3	43.0	37.4
	DBStd	12.98	8.48	7.97	6.85	6.95	6.57	6.11	6.02	5.60	5.52	6.50	6.57	8.29
	HDD50	1861	450	366	260	91	13	0	0	0	2	64	223	392
	HDD65	5504	913	784	714	459	270	125	51	54	193	427	660	854
	CDD50	2091	1	2	11	82	221	368	516	497	275	102	13	3
	CDD65	259	0	0	0	0	12	43	102	86	16	0	0	0
	CDH74	2429	0	0	0	17	149	413	964	751	130	5	0	0
	CDH80	760	0	0	0	1	27	111	336	259	26	0	0	0
Wind	WSAvg	8.7	10.3	9.9	9.8	8.3	7.8	7.8	7.7	7.4	7.7	8.5	8.9	9.9
Precipitation	PrecAvg	30.9	2.8	2.2	2.0	1.9	2.6	2.6	3.2	3.1	2.6	2.6	2.7	2.8
	PrecMax	41.5	4.9	4.8	5.0	3.6	6.3	4.6	7.3	7.3	5.8	7.3	5.2	6.7
	PrecMin	23.6	0.6	0.7	0.4	0.3	0.9	1.0	1.2	0.9	0.7	0.8	0.3	0.9
	PrecStd	4.6	1.3	1.1	1.2	0.8	1.3	0.9	1.4	1.4	1.3	1.3	1.2	1.2
Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	55.4	57.5	66.0	76.7	83.8	89.2	93.2	91.8	83.9	72.6	60.5	55.8
		MCWB	49.9	48.0	51.8	59.0	65.2	69.6	68.5	68.7	66.3	61.6	55.5	51.1
	2%	DB	51.9	53.5	60.1	71.1	78.3	82.8	87.9	86.8	77.6	66.8	57.2	53.5
		MCWB	47.9	47.2	50.0	56.2	62.3	66.2	67.4	66.5	63.7	59.1	52.7	49.7
	5%	DB	48.9	50.4	55.6	66.6	73.7	78.7	83.6	82.1	73.0	63.2	55.0	50.4
		MCWB	45.8	45.9	48.2	54.0	59.4	64.0	66.1	65.4	61.7	57.2	51.0	46.9
	10%	DB	46.4	48.1	52.2	62.5	69.8	74.8	79.1	77.3	68.8	60.6	52.1	48.2
		MCWB	43.6	44.2	46.4	52.4	57.4	62.3	64.5	63.6	59.9	55.9	48.4	45.3
Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	51.5	50.4	54.2	60.4	67.8	70.8	71.2	70.7	67.3	62.4	56.1	52.3
		MCDB	54.2	54.3	62.0	72.0	80.7	84.1	85.3	85.4	79.2	70.3	58.9	55.4
	2%	WB	48.6	48.4	51.4	57.5	63.8	67.6	69.2	68.5	65.1	59.9	53.3	50.1
		MCDB	51.6	52.4	57.8	67.9	74.9	79.7	83.2	82.0	75.2	65.8	56.6	53.0
	5%	WB	45.9	46.2	49.2	55.3	61.0	65.5	67.6	66.8	62.8	57.9	51.2	47.5
		MCDB	48.6	50.1	54.5	64.2	71.5	76.3	80.2	78.4	70.6	62.5	54.6	50.3
	10%	WB	43.6	44.0	46.9	53.3	58.6	63.3	65.7	65.1	60.9	56.0	49.1	45.2
		MCDB	46.1	47.7	51.8	61.7	67.3	72.4	76.7	75.1	67.8	60.1	51.8	47.7
Mean Daily Temperature Range	5% DB	MDBR	8.3	10.1	13.2	17.0	18.1	17.9	18.0	17.9	15.8	12.9	8.9	7.8
		MCDBR	10.4	13.2	20.0	24.4	25.1	24.9	25.7	25.9	22.5	16.2	11.3	9.6
		MCWBR	8.5	9.2	12.1	12.6	12.4	11.5	10.1	10.4	11.2	10.1	8.4	8.1
	5% WB	MCDBR	9.5	11.4	17.3	21.3	22.4	22.1	22.6	21.4	19.4	15.3	10.6	9.1
		MCWBR	8.3	8.6	11.2	11.9	12.0	11.4	9.8	9.9	10.5	10.1	8.4	8.1
Clear-Sky Solar Irradiance	taub	0.319	0.340	0.374	0.411	0.412	0.410	0.425	0.418	0.387	0.357	0.337	0.314	
	taud	2.364	2.351	2.272	2.214	2.247	2.272	2.241	2.276	2.355	2.420	2.392	2.355	
	Ebn at Noon	213	243	256	260	266	267	260	255	250	234	203	192	
	Edh at Noon	19	26	34	40	41	40	41	37	31	24	19	16	
All-Sky Solar Radiation	RadAvg	201	419	755	1247	1525	1630	1560	1309	954	548	246	157	
	RadStd	20	60	87	149	164	124	206	99	118	52	28	15	

Historical Trends

		DBAvg	Heating		Cooling			Degree-Days			
			99% DB	99% DP	1% DB	1% WB	1% DP	HDD50	HDD65	CDD50	CDD65
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)
(46) Station Only		+0.72	N/A	N/A	N/A	N/A	+0.86	N/A	-288	N/A	N/A
(47) Regional (46 neighbors)		N/A	N/A	N/A	N/A	N/A	N/A	N/A	-223	N/A	N/A

Nomenclature: See separate page